Critical Analysis of CRPD Alternative Reports: A Computational Approach to Disability Rights Monitoring

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Abstract

Assessment of CRPD implementation relies primarily on State Party self-reports, which often emphasize formal commitments over lived realities. Civil society alternative reports are intended to counterbalance this limitation by documenting implementation gaps and on-the-ground experiences of persons with disabilities, yet they remain largely unexamined in research. This paper will apply computational text analysis to a corpus of civil society alternative reports submitted to the CRPD. We aimed to examine policy priorities, implementation concerns, and reporting patterns across regions and institutional contexts. By centering shadow perspectives at scale, the study aims to highlight how alternative reports reveal dimensions of disability inclusion that are obscured in official reporting and demonstrate the value of triangulated, multi-informant approaches for evaluating progress under the CRPD.

Keywords: CRPD implementation, Disability rights monitoring, Civil society alternative reports, Computational text analysis, Policy accountability

1. Introduction

State Reports have become the main evidence base for assessing global progress on disability inclusion (Cogburn et al. (2025), Subic & Ferri (2022)). Over 16% of the global population lives with disabilities, with more than 80% residing in the Global South (WHO, 2024; UNDRR, 2023). The United Nations Convention on the Rights of Persons with Disabilities (CRPD), ratified by 193 countries, is the primary international framework protecting disability rights. As a binding treaty, it requires State Parties to submit periodic reports outlining their implementation efforts, and these State Reports have become the main evidence base for assessing global progress on disability inclusion. In parallel, civil society organizations submit alternative reports to the CRPD Committee, offering independent assessments of how countries are implementing their obligations and highlighting gaps that may not appear in official accounts.

Despite their formal recognition, alternative reports remain largely absent from scholarly analysis, which continues to rely predominantly on State reporting (see Cogburn et al. (2025); Subic & Ferri (2022); Chaney (2021)). This creates a significant knowledge gap in disability rights research and policy evaluation. While State reports reflect legislative commitments and institutional reforms, they may not fully capture lived experiences, systemic discrimination, or implementation failures. A systematic analysis of alternative reports is therefore essential to strengthen accountability, identify gaps between policy and practice, and center the perspectives of disability communities in global treaty monitoring.

2. Literature Review

Civil Society and Human Rights Treaty

Civil society participation in international human rights monitoring is widely recognized as central to accountability and transparency. Gaer (1995) describes non-governmental organizations as the eyes and ears of UN treaty bodies, providing information that governments may omit or distort. In the disability rights context, this role is even more significant. Fina et al. (2017) characterizes the CRPD as marking a participatory turn in human rights law, positioning persons with disabilities and their representative organizations not merely as stakeholders but as active contributors to monitoring and implementation. Yet despite this recognized importance, the actual content, framing, and influence of alternative reports remain understudied. Systematic analysis of what these reports emphasize and how they construct disability rights issues is largely absent from the literature, a notable gap given the hundreds of civil society submissions archived across reporting cycles in the OHCHR database.

Pitfalls of State Self-Reporting

Scholarships have questioned the reliability of state self-reporting as an indicator of genuine treaty compliance. Simmons (2009) argues that ratifying human rights treaties and submitting periodic reports does not automatically translate

into improved human rights outcomes, particularly where domestic accountability mechanisms are weak. In the CRPD context, similar concerns have emerged. Oyaro (2015) finds that several African state reports emphasized legislative reform while overlooking continued reliance on charity-based approaches and limited institutional change. Subic & Ferri (2022) observes that European disability strategies often adopt inclusive language without embedding meaningful implementation mechanisms, and Cogburn et al. (2025) explicitly acknowledge that reliance on self-reported CRPD state reports constrains assessments of governments' true policy priorities and real-world impact. Together, these findings suggest that State Reports may present an overly optimistic picture of CRPD implementation.

Computational Analysis in Policy Research

Recent advances in Natural Language Processing (NLP) and large language models (LLMs) offer powerful tools for analyzing large-scale policy texts. Cogburn et al. (2025) demonstrate that hybrid approaches combining traditional text mining with generative AI can systematically examine CRPD State Reports, using term frequency analysis, topic modeling, named entity recognition, and dictionary-based categorization to identify policy priorities, regional variation, and shifts from medical to rights-based paradigms. Similar methods have been applied elsewhere: B. T. Smith et al. (2021) use document embeddings to analyze UN Sustainable Development Goal reports, and Törnberg (2023) explore the integration of LLMs into qualitative text analysis while cautioning about bias, hallucination, and interpretability limits. These computational approaches are particularly well suited to alternative reports, given their volume, multilingual nature, and the need for cross-national and longitudinal comparison. Yet despite these methodological developments, no existing research has systematically applied such techniques to the corpus of CRPD alternative reports.

3. Current Focus

The disability research literature consistently shows that self-assessment frameworks are prone to systematic bias, including underestimation of problem severity (Pole et al., 2006), over-representation of success cases (Gotham et al., 2015), under-reporting among populations with communication barriers (Elkasabi, 2021), and divergence from external observation, particularly in domains involving social functioning and subjective experience (Pearl et al., 2016; Sandercock et al., 2020). Established methodological principles such as triangulation (Brown, 2001), multi-informant assessment (Sandercock et al., 2020), cross-source validation procedures (Fujiura, 2012), and the integration of multiple data streams for robust research design (Iezzoni, 2002) all point to the same conclusion: privileging State Party self-reports while marginalizing civil society alternative reports misunderstands their complementary epistemic roles. Empirical evidence further demonstrates that self-reports and alternative reports capture different dimensions of disability experience (Lee et al., 2004), rely on distinct information sources (E. Smith et al., 2025), apply different severity calibrations to similar conditions (Fujiura, 2012), and diverge most sharply in areas central to the CRPD's transformative vision, including quality of life, social inclusion, unmet needs, and subjective well-being (Sandercock et al., 2020). Computational analysis of alternative and State reports together enables systematic identification of these divergences, quantification of magnitude gaps, among others, operationalizing at scale the validation principles disability research identifies as essential for reliable and meaningful CRPD monitoring.

4. Conceptual Framework

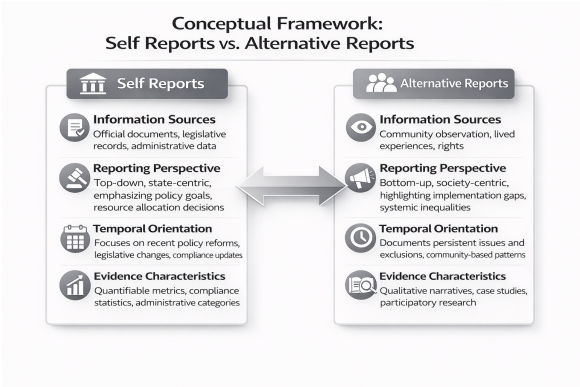

Figure 1: Conceptual framework for CRPD monitoring

5. Research Questions

The primary research questions that guide this study are as follows:

RQ1: What policy priorities and implementation concerns do civil society organizations emphasize in CRPD alternative reports?

RQ2: How do alternative reports vary across regional, economic, and institutional contexts?

RQ3: What actors, institutions, evidence types, and demographic groups do civil society organizations centre in their reporting?

The following questions builds on the primary research questions:

RQ1: Policy Priorities

RQ1.1: What are the most frequently occurring terms across alternative reports (Term Frequency)?

RQ1.2: Which terms are most distinctive or context-specific across reports (TF-IDF)?

RQ1.3: What key phrases and advocacy themes emerge from the reports (N-grams)?

RQ1.4: How do results differ across term frequency, TF-IDF, and N-gram analyses?

RQ2: Regional and Contextual Variation

RQ2.1: How do policy priorities vary across geographic regions (TF-IDF comparison)?

RQ2.2: How does report length and level of detail differ across regions?

RQ2.3: How does lexical diversity vary across regions (Type-Token Ratio)?

RQ3: Actors, Institutions, Evidence, and Demographic Groups

RQ3.1: Which actors and institutions are most frequently referenced in the reports (Named Entity Recognition)?

RQ3.2: What types of evidence are used to support claims (dictionary-based classification)?

RQ3.3: What is the sentiment or evaluative tone of the reports (lexicon-based sentiment analysis)?

RQ3.4: Which specific demographic groups are identified as facing the most severe intersectional discrimination? (NotebookLM / LLM-based qualitative extraction)

6. Methodology

Each research question is addressed using a distinct set of computational and qualitative analytical techniques. RQ1 is examined through frequency-based text analysis (TF, TF-IDF, and N-grams). RQ2 is addressed using comparative regional analysis, including TF-IDF, document length, and lexical diversity measures. RQ3 is analyzed using named entity recognition (NER), dictionary-based classification, lexicon-based sentiment analysis, and qualitative large language model-assisted coding to identify intersectional demographic groups.

6.1 Data Collection and Preprocessing

The CRPD Alternative Reports were extracted from the official UN Treaty Body Database maintained by the Office of the High Commissioner for Human Rights (OHCHR). Data collection was conducted programmatically using Python with Selenium automation, applying filters for Committee (CRPD), document type ("Info from Civil Society Organizations (for LOIs)"), and English-language submissions. Documents were downloaded in. A total of 337 alternative reports were collected.

The data preparation consisted of the following steps:

(1) Imported the raw documents into Positron using the `readtext` package and converted them into a dataframe (`corpus_df`) containing `doc_id` and text.

(2) Tokenized the corpus using `unnest_tokens()` from the tidytext package, producing word-level tokens stored in `alt_reports_words`.

(3) Removed stopwords in multiple stages, including default English stopwords, CRPD-specific terms, legal-bureaucratic terms, and methodological artifacts, combining them into a unified stopword dictionary.

(4) Removed document-structure artifacts such as Table of Contents markers and numeric page references.

(5) Generated a cleaned dataset (`alt_reports_cleaned`) containing meaningful tokens across 337 reports.

(6) Constructed a Document–Term Matrix (DTM) using term-frequency weighting and removed sparse terms with a 0.99 sparsity threshold.

(7) Extracted metadata from filenames to generate variables for country, continent, and region, enabling comparative regional analysis.

Term Frequency, TF-IDF, and N-gram Analysis

To address RQ1.1-RQ1.4, we employ multiple frequency-based text analysis techniques. Term Frequency (TF) is used to identify the most commonly occurring terms across the corpus, providing a baseline measure of core policy priorities. TF-IDF (Term Frequency–Inverse Document Frequency) is used to identify distinctive terms that are highly specific to individual reports or regions, allowing the detection of localized policy concerns. N-gram analysis (specifically bigrams) is used to extract frequently co-occurring word pairs, enabling identification of structured policy themes such as "mental health" and "legal capacity." Comparative analysis across TF, TF-IDF, and N-grams is conducted to assess how different methods capture overlapping versus distinct dimensions of civil society advocacy. In addition, we implement GloVe-based word embeddings to capture semantic relationships between key concepts and apply custom dictionary matching to group terms into policy-relevant thematic categories.

RQ2: Regional and Contextual Comparative Analysis

To address RQ2.1–RQ2.3, we conduct a grouped comparative analysis across geographic regions. TF-IDF is applied at the regional level to identify "policy signatures" that

distinguish different regions. Report length and document-level token counts are used to assess variation in reporting depth and advocacy capacity. Lexical diversity is measured using the Type-Token Ratio (TTR), capturing variation in conceptual richness across reports. Metadata integration enables classification into seven global regions, allowing systematic cross-regional comparison.

RQ3: Actors, Institutions, Evidence, and Sentiment Analysis

To address RQ3.1–RQ3.3, we employ a combination of natural language processing and dictionary-based methods. Named Entity Recognition (NER) is used to identify key actors and institutions referenced in the reports, including governmental bodies, international organizations, and Disabled Persons' Organizations (DPOs).Dictionary-based classification is applied to categorize types of evidence used in the reports, including statistical data, surveys, legal references, and qualitative case descriptions. Lexicon-based sentiment analysis is conducted using established sentiment dictionaries to assess evaluative tone, identifying patterns of critique, concern, or positive framing in relation to CRPD implementation.

RQ3.4: Intersectional Demographic Group Identification Using LLM-Assisted Qualitative Analysis

To address RQ3.4, we implement a structured qualitative analysis using NotebookLM as a large language model-assisted research tool. The full corpus of CRPD alternative reports is provided to NotebookLM, and a standardized analytical prompt is used to guide extraction of intersectional demographic groups. The prompt instructs the model to; identify all demographic groups referenced in the reports (e.g., women with disabilities, children with disabilities, indigenous persons with disabilities), detect intersectional combinations of identity (e.g., gender + disability + poverty), extract language indicating severity (e.g., "most marginalized," "particularly vulnerable"), normalize terminology across reports and rank groups based on frequency, severity of language, and number of intersecting vulnerability factors

The model produces a structured output consisting of; Ranked demographic groups, explanations of vulnerability, key intersectional factors, illustrative excerpts or paraphrased evidence and regional or contextual variation. To ensure analytical rigor, this approach is treated as a qualitative coding layer rather than an autonomous inference system. The prompt constrains the model to base all outputs strictly on the provided corpus, minimizing speculative interpretation. The resulting outputs are used to identify patterns of intersectional discrimination across reports and to complement the quantitative findings from TF-IDF and dictionary-based analysis.

Methodological Integration

This study adopts a mixed-methods computational approach that integrates; Frequency-based text analysis (TF, TF-IDF, N-grams), semantic modeling (GloVe embeddings),dictionary-based classification, named entity recognition, lexicon-based sentiment analysis and LLM-assisted qualitative coding (NotebookLM). By combining these approaches, the methodology enables both large-scale pattern detection and nuanced identification of intersectional vulnerability, providing a comprehensive framework for evaluating civil society perspectives on CRPD implementation.

7. Findings

The findings are organized according to the three primary research questions. Section 7.1 addresses RQ1 on overall policy priorities and implementation concerns, Section 7.2 addresses RQ2 on regional and contextual variation, and Section 7.3 addresses RQ3 on actors, institutions, evidence types, sentiment, and intersectional demographic groups. Each sub-section reports the result of a specific computational technique and explicitly links back to the secondary research question it answers.

7.1 Policy Priorities and Implementation Concerns

Core Policy Priorities

The analysis identified the most commonly occurring substantive terms across the 337 alternative reports. The ten most frequent terms were people, children, services, education, health, government, social, law, support, and national, indicating that civil society reporting is anchored in core social-rights domains covering populations, public systems, and the institutions responsible for delivery. The association analysis strengthened this interpretation: testing correlations of services against the Document-Term Matrix at a 0.99 sparsity threshold revealed statistically meaningful links with personal ($r = 0.78$), assistance ($r = 0.74$), lack ($r = 0.73$), and access ($r = 0.72$). The pairing with personal and assistance signals a focus on independent-living support, while the correlations with lack and access confirm that civil society frames service provision through a vocabulary of deprivation rather than progress, establishing a diagnostic rather than celebratory orientation from the outset.

Distinctive and Context-Specific Terms

The document-level TF-IDF analysis was applied to surface terms that are highly frequent in a specific report yet rare across the corpus, isolating the unique policy signature of each submission. The highest TF-IDF scores revealed a striking thematic cluster around leprosy in reports from Laos, Indonesia, Mexico, and Venezuela, accompanied by terms such as stigmatisation and infectious, indicating that civil

society in these contexts is centring the rights of persons affected by neglected tropical diseases. Other distinctive terms included intersex in the Netherlands, albinism in Angola and Malawi, Roma in Moldova, Khmer Krom in Vietnam, and Åland in Finland. Report-level visualisations further confirmed localised emphases, including LGBTI and Tbilisi in Georgia and the extradition debate in Hong Kong. Alternative reports are therefore not standardised replicas of a global CRPD template; they are targeted, context-specific diagnostics that surface marginalised populations and local crises that universal rights vocabulary would otherwise hide.

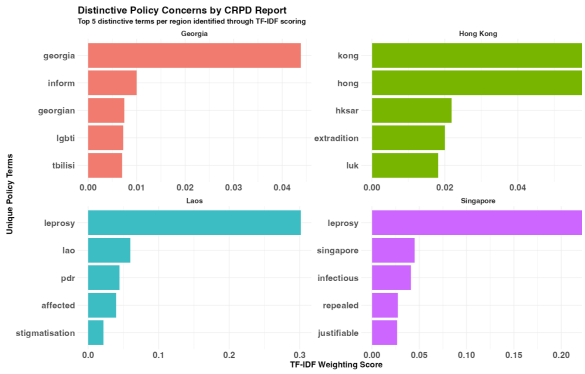

Figure 2: Distinctive Policy Concerns by CRPD Report

Policy Phrases and Advocacy Themes

The bigram analysis shifted focus from isolated terms to structured policy instruments. The top bigrams across the corpus were mental health (n = 2,133), disabled people (n = 1,693), legal capacity (n = 1,546), sign language (n = 1,315), corporal punishment (n = 1,305), reasonable accommodation (n = 1,080), inclusive education (n = 1,018), provide information, health care, and social protection. These pairings correspond directly to specific CRPD articles. Legal capacity and reasonable accommodation point to Articles 12 and 5, while inclusive education, sign language, and corporal punishment map onto Articles 24, 21, and 16 respectively. The prominence of mental health and legal capacity reflects a coordinated global emphasis on psychosocial disability and the dismantling of substituted decision-making regimes. The high frequency of corporal punishment suggests that civil society is using the CRPD reporting process as a channel for flagging severe violations against children with disabilities, including violations that overlap with obligations under the Convention on the Rights of the Child.

A complementary custom-dictionary analysis grouped stem variants into three consolidated policy categories (inclusive education, health services, and employment access) and confirmed a thematic imbalance across reports. Individual reports exhibited highly uneven footprints: certain reports contained over 140 mentions of inclusive-education terms while others concentrated on health services, indicating that civil society organisations tailor their advocacy to the specific sector where their national State Party is perceived to be failing.

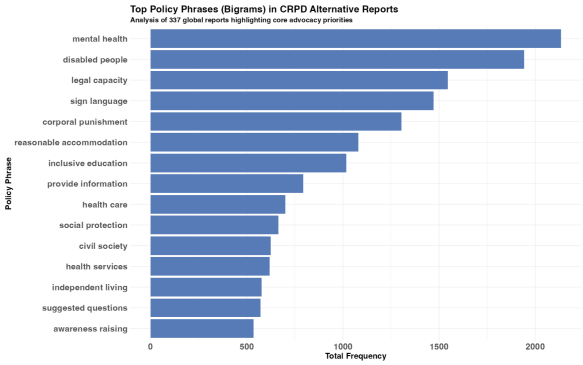

Figure 3: Bi-grams Analysis

Cross-Method Synthesis and Semantic Structure

The comparison of three methods shows that each captures a different dimension of advocacy. Term frequency identifies the broad agenda (services, education, health), bigrams reveal the specific policy instruments used to articulate that agenda (mental health, legal capacity, reasonable accommodation), and TF-IDF surfaces the localised crises that neither of the first two methods can detect (leprosy, albinism, intersex rights, ethnic minorities). The three methods are complementary rather than redundant, and relying on any single method would produce a distorted picture of civil society priorities.

The GloVe word-embedding model added a further semantic layer by mapping conceptual proximity rather than co-occurrence frequency. The ten words most semantically similar to accessibility included requirements (cosine similarity = 0.76), standards (0.72), transport (0.62), and public (0.61), alongside guidelines and systems. This pattern indicates that civil society does not treat accessibility as an abstract principle but as a regulatory obligation tied to formal technical standards, particularly in transport and public infrastructure. The semantic map therefore reinforces the overall finding from the frequency analyses: alternative reports operationalise CRPD rights as concrete, measurable state obligations rather than aspirational ideals.

7.2 Regional and Contextual Variation

Regional TF-IDF Signatures

Applying TF-IDF at the regional level produced a clear set of regional policy signatures. Europe and Central Asia

centred on mature rights-framework terminology, including legal capacity, deinstitutionalisation, independent living, and mental health reform. Sub-Saharan Africa highlighted albinism, community-based rehabilitation, and violence, reflecting advocacy shaped by targeted attacks on persons with albinism and under-developed service infrastructure. East Asia and the Pacific emphasised accessibility, sign language, and leprosy, the last indicating the continued human-rights dimensions of neglected tropical diseases in the region. The Middle East and North Africa centred on guardianship, women with disabilities, and Gaza strip, the latter reflecting conflict-related disability concerns that are absent from other regions. Latin America and the Caribbean prioritised participation, violence, and intersectionality, while South Asia and North America produced too few reports to generate stable regional signatures. These results confirm that although the CRPD is a universal treaty, the implementation gaps reported by civil society are deeply structured by regional political, legal, and social contexts.

A complementary regional bigram analysis, restricted to the four highest-reporting regions, confirms and extends the TF-IDF pattern (Figure 4). The most frequent policy phrases show both a shared core and sharp regional divergence. Mental health, legal capacity, corporal punishment, and reasonable accommodation appear as top phrases in multiple regions, indicating a set of universal advocacy pillars consistent with the bigram findings in RQ1. Beyond this shared core, each region foregrounds its own priorities. East Asia and the Pacific is the only region where sign language is the single most frequent phrase, consistent with regional advocacy around deaf community rights and linguistic recognition. Europe and Central Asia is anchored by disabled people, mental health, legal capacity, and independent living, reflecting mature deinstitutionalisation and autonomy-focused advocacy. The Middle East and North Africa is the only region in which Gaza strip appears in the top five phrases, reflecting conflict-related disability concerns entirely absent from other regional corpora. Sub-Saharan Africa prioritises mental health, legal capacity, and corporal punishment, with South Africa entering the top five because of the disproportionate volume of reports produced by South African civil society within the regional total. The bigram frequencies therefore triangulate the TF-IDF findings: the regions do not merely use different rare terms, they also organise their most common advocacy language around different policy problems.

Report Length and Reporting Depth

Regional variation was also evident in the length of individual reports. Word-count analysis showed that Europe and Central Asia produced the longest reports on average, frequently exceeding 3,800 words per submission, with high

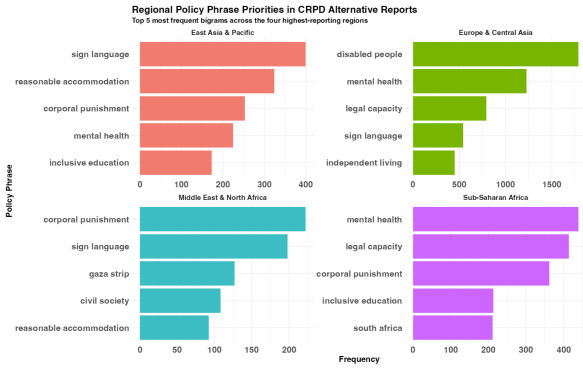

Figure 4: Regional Policy Phrases

variance indicating a range of both short focused submissions and extended thematic assessments. East Asia and the Pacific and Latin America and the Caribbean produced moderately long reports, while Sub-Saharan Africa, the Middle East and North Africa, and South Asia produced shorter reports on average. North America produced the fewest and shortest reports overall. The volume of reports also varied sharply, with Europe and Central Asia contributing approximately 150 of the 337 reports (roughly 45 percent of the corpus), followed by East Asia and the Pacific and Sub-Saharan Africa in a second tier, and South Asia and North America producing the fewest. Taken together, these patterns indicate that advocacy capacity, measured both in quantity of submissions and in depth per submission, is strongly concentrated in regions with dense NGO networks, stable treaty-body engagement, and developed disability-rights institutions.

Lexical Diversity Across Regions

The Type-Token Ratio (TTR), computed per document and then averaged by region after filtering to reports with at least 100 tokens. TTR values were broadly similar across regions, with mean ratios clustering between 0.15 and 0.30, but meaningful ordering emerged. Regions producing shorter reports tended to display higher TTR values because their limited vocabulary is distributed across fewer total tokens, while regions producing longer and more elaborated reports displayed lower TTR values driven by repetition of core policy vocabulary. The lexical-diversity pattern therefore complements the length analysis: Europe and Central Asia produced longer but more lexically repetitive reports, reflecting the standardised vocabulary of mature advocacy frameworks, while some smaller regional corpora showed higher lexical diversity relative to their size. This suggests that shorter reports are not necessarily less substantive; they may concentrate on targeted issues using a broader conceptual vocabulary rather than repeating standardised framework language.

Regional Contexts and Localized Crises

While the CRPD is a universal framework, our analysis proves that the implementation gaps reported by civil society are deeply shaped by specific political, geographic, and economic contexts. Europe & Central Asia contributed nearly half of all reports (150), reflecting a dense network of NGOs and strong institutional capacity. In contrast, regions like South Asia and North America contributed the fewest.

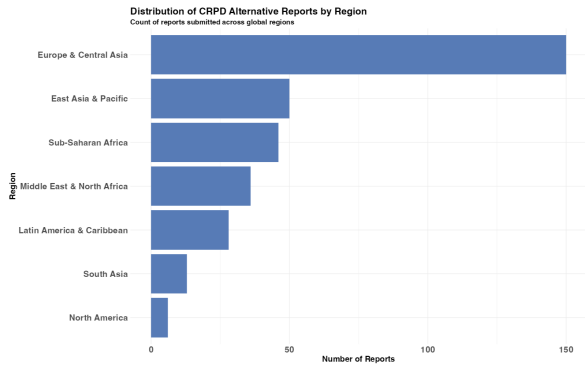

Figure 5: Regional Participation

7.3 Actors, Institutions, Evidence, Sentiment, and Demographic Groups

Actors and Institutions

Named Entity Recognition performed through spaCy, followed by multi-word entity consolidation, identified the most frequently referenced institutions and persons across the corpus. Among organisations, the most frequently cited were the European Union (n = 180), Disabled Persons' Organisations (DPO, n = 162), the Supreme Court (n = 98), the European Commission (n = 81), the European Court (n = 54), and the National Council (n = 44). This hierarchy shows that civil society deliberately addresses its reporting to high-level legislative and judicial power centres rather than lower-level administrative bodies. The prominence of DPO as a top institutional reference is equally important: civil society positions its own representative organisations as authoritative voices alongside formal state and supranational institutions, consistent with the CRPD's nothing about us without us principle.

The person-entity results surfaced primarily national identifiers rather than named individuals, with Malawi (n = 36) and Jordan (n = 19) appearing most frequently in this category. These results reflect two tendencies. First, the NER model at times classifies country or regional names as persons when used as proxies for national advocates. Second, civil society reports generally prioritise collective national voices and documented evidence over profiling of individual figures. The frequent appearance of https (n = 128) and hyperlink (n = 20) as entities further indicates that alternative reports rely heavily on external web-based documentation, including linked reports, press materials, and official sources, as part of their evidentiary strategy.

Evidence Types

The dictionary-based evidence classification measured the relative frequency of four evidence categories: statistical data, surveys and polls, academic research, and case studies. Statistical data was overwhelmingly dominant, with approximately 5,000 references across the corpus, followed at a substantial distance by surveys and polls and academic research, while case studies were the least frequently used category. This pattern demonstrates that civil society has adopted a quantitative, data-first approach to treaty-body advocacy, grounding claims of State non-compliance in census data, percentages, and formal datasets rather than in individual anecdotes. The relatively low count for case studies does not indicate an absence of lived experience in the reports; rather, it reflects that personal testimony is typically embedded within broader statistical and institutional critiques rather than presented as a standalone evidentiary genre. The dominance of statistical evidence signals a deliberate professionalisation of shadow reporting, positioning civil society as a rigorous empirical counterweight to State self-assessment.

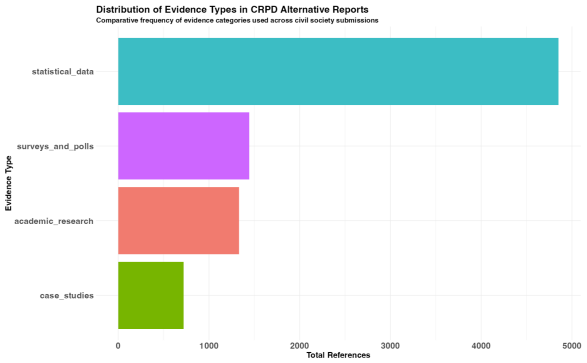

Figure 6: Evidence Types

Sentiment Analysis

The Bing dictionary classified tokens in the cleaned corpus as positive or negative. The overall distribution was skewed strongly toward negative sentiment, with negative tokens outnumbering positive tokens across the corpus as a whole. This critical posture is consistent with the strategic role of alternative reports: civil society organisations are not writing balanced performance reviews but accountability documents intended to surface implementation gaps, systemic barriers, and rights violations. The combination of negative sentiment with the "data-first" evidence pattern

identified in Section 7.3.2 is particularly important. It indicates that civil society critique is not simply rhetorical or emotive but is explicitly grounded in statistical documentation, making the negative tone a structured diagnostic tool rather than a matter of advocacy style.

Intersectional Demographic Groups

The LLM-assisted qualitative analysis conducted in NotebookLM identified and ranked demographic groups described in the reports as facing the most severe forms of intersectional discrimination. The resulting ranked list, from highest to lowest severity, was: (1) women and girls with disabilities, (2) children and youth with disabilities, particularly those in institutional settings, (3) migrants, refugees, asylum seekers, and internally displaced persons with disabilities, (4) persons with psychosocial and intellectual disabilities, (5) Indigenous peoples, ethnic minorities including Roma, and persons with albinism with disabilities, and (6) LGBTI and intersex persons with disabilities.

Four intersectional patterns recurred across the corpus: gender combined with disability and legal disempowerment; disability combined with age and either poverty or cultural stigma; displacement combined with disability and exposure to violence; and ethnicity or race combined with disability and rural isolation. Regional variation was clear. In Global South contexts, intersectional discrimination was most often tied to poverty, cultural stigma, and infrastructure gaps, producing extreme exclusion and violence. In Global North contexts, discrimination was more institutionalised, operating through legal, medical, and bureaucratic systems such as substituted decision-making regimes and coercive medical practices. In conflict and migration settings, risks were immediate and life-threatening, while in stable environments they manifested as long-term structural exclusion from services, employment, and political participation. These qualitative findings complement the quantitative TF-IDF regional signatures by identifying who is affected by the localised crises detected in the frequency-based analyses.

8. Discussion

The findings collectively show that CRPD alternative reports function as a pivotal genre of human-rights documentation on State reports. Three interpretive points emerge from the analysis.

First, the combination of term frequency, TF-IDF, and bigram results shows that civil society operates simultaneously at two scales: a universal scale grounded in shared CRPD vocabulary (mental health, legal capacity, inclusive education, reasonable accommodation) and a localised scale that surfaces issues invisible in universalising frameworks (leprosy, albinism, intersex rights, ethnic-minority inclusion, conflict-affected populations). Treaty-monitoring approaches that privilege only the universal scale systematically under-report the specific populations the CRPD was designed to protect. The two scales are not in tension; they are two analytical registers civil society actively deploys to diagnose the gap between treaty ratification and lived implementation.

Second, the regional analysis indicates that the participatory turn described by Fina, Palmisano, and Cera (2017) is real but structurally uneven. The near-half share of reports produced by Europe and Central Asia, combined with the longer average report length in that region, shows that advocacy capacity is strongly concentrated in well-resourced regions with mature civil-society infrastructure. Regions with smaller corpora are not less affected by disability-rights violations; they are less able to document those violations at the scale required for meaningful treaty-body engagement. This imbalance has direct methodological consequences. Any analysis that weights findings purely by frequency, including aggregate global priorities derived from term-frequency counts, will inherit this regional skew and report European priorities as though they were global ones.

Third, the evidence, sentiment, and institutional findings taken together indicate a clear professionalisation of shadow reporting. Civil society is not operating as a collection of isolated advocacy groups submitting anecdotal accounts. The overwhelming reliance on statistical evidence, the sustained negative evaluative tone, the strategic targeting of high-level institutions such as the European Union and supreme courts, and the centrality of DPOs as authoritative voices all indicate a coordinated, data-driven accountability practice. This is consistent with the broader argument of Cogburn et al. (2025) that NLP methods are suited to CRPD analysis precisely because the documents themselves are becoming more structured and data-rich.

The LLM-assisted intersectional analysis also has implications for how treaty bodies interpret alternative reports. Frequency-based methods consistently surface groups that appear often in the text (women with disabilities, children), but the most severe intersectional vulnerabilities often involve compounded identities (displaced persons with psychosocial disabilities, Indigenous women with disabilities in rural areas) that may not register in simple term counts. Qualitative coding, whether conducted manually or through LLM assistance, therefore remains essential for capturing the full picture of who is most marginalised under the Convention.

9. Policy Implications

The CRPD Committee should more systematically integrate alternative reports into its review process through computational summaries and cross-report comparisons. Governments should also treat divergences between State and alternative reports as signals of genuine implementation gaps rather than mere inconsistencies. Policymakers must incorporate civil society evidence into evaluation frameworks and strengthen participation channels for disability communities, recognizing that recurring themes such as lack of access, weak service delivery, and unmet needs reflect real failures in translating formal commitments into lived realities.

Civil society organizations should adopt more standardized, data-driven reporting practices to enhance cross-country comparability and amplify their influence in global monitoring. This study further demonstrates that combining traditional NLP methods with LLM-assisted analysis yields the most comprehensive insights: TF-IDF and N-grams surface structured patterns, while LLMs capture nuanced intersectional dynamics that frequency-based methods alone cannot detect. LLM outputs must nonetheless be treated as a qualitative complement rather than a standalone inference system, with careful attention to bias and interpretability. This has broader implications for the responsible use of AI in policy research and human rights monitoring.

10. Recommendations

Based on the findings and discussion, we offer four recommendations for the CRPD Committee, funders, DPOs, and the broader disability rights research community.

First, the CRPD Committee should integrate computational analysis of alternative reports into its List of Issues Prior to Reporting (LOIPR) procedures. Regional TF-IDF signatures show that distinctive local concerns, including leprosy, albinism, and conflict-related disability issues, are consistently raised by civil society but unlikely to appear in State self-reports. Structured computational review prior to LOIPR drafting would allow Country Rapporteurs to ask more targeted, locally grounded questions.

Second, international funders and human rights donors should invest in the documentation capacity of DPOs in under-resourced regions. The correlation between reporting volume and civil society infrastructure means the current corpus systematically under-represents South Asia, North America, and parts of Sub-Saharan Africa and the Middle East. Targeted funding for report preparation, data collection, and translation would reduce the structural bias that currently weights the global evidence base toward European priorities.

Third, the Committee should adopt locally nuanced review frameworks rather than uniform evaluative grids. Civil society priorities in Sub-Saharan Africa, centered on albinism, community-based rehabilitation, and violence, differ substantially from those in Europe and Central Asia, which focus on legal capacity, deinstitutionalization, and mental health reform. Standardized grids risk misreading legitimate regional differences as weak advocacy.

Fourth, future research should expand the corpus to include non-English submissions, pair alternative reports with corresponding State reports for direct comparative analysis, and apply longitudinal methods across reporting cycles to test whether civil society critique produces measurable shifts in State Party compliance language over time.

11. Future Research

Future research should extend this framework in three directions. First, the corpus should be expanded to include non-English submissions to reduce the current Anglophone bias. Second, alternative reports should be paired with corresponding State reports to enable direct quantitative comparison of lexical choices, sentiment distributions, and thematic priorities across informant types. Third, longitudinal analysis across reporting cycles would allow researchers to detect whether civil society critique produces measurable shifts in State Party language over time, offering a concrete test of whether shadow reporting exerts observable pressure on treaty compliance.

12. Limitations

This study has four principal limitations. First, the corpus is geographically skewed, with Europe and Central Asia contributing nearly half of all reports. Global aggregates, including the term-frequency, bigram, and sentiment results, are therefore weighted toward European institutional priorities and should not be read as strictly representative of civil society advocacy worldwide. Second, the corpus is restricted to English-language submissions, which excludes a substantial portion of alternative reports produced in other UN working languages and in national languages. This restriction almost certainly under-represents localised advocacy that never enters the English-language OHCHR pipeline. Third, the dataset lacks granular metadata on State Party income classification, political regime type, and reporting cycle, which prevented formal economic and temporal modelling of how national wealth and regime characteristics correlate with specific advocacy priorities. Fourth, the NER and dictionary-based components rely on pre-trained models and researcher-defined categories. Although, this is a stan-

dard practice and it introduces the possibility that important actors or evidence types that do not match model training data or dictionary entries are under-counted. implementation gaps.

13. Conclusion

This study applied a mixed-methods computational framework to 337 civil society alternative reports submitted to the CRPD Committee, combining frequency analysis, TF-IDF, bigrams, semantic embeddings, regional comparison, Named Entity Recognition, sentiment analysis, and LLM-assisted qualitative coding. The results show that alternative reports are independent accountability documents that pair universal CRPD vocabulary with highly localised diagnostics, surface populations routinely obscured in State self-reporting, rely predominantly on statistical evidence, sustain a critical evaluative tone, and strategically target high-level institutional actors. These patterns have a clear implication for CRPD monitoring: evidence derived from State reports alone is systematically incomplete, and centring civil society reporting at scale offers a more transparent foundation for evaluating disability rights under the Convention. The framework is reproducible and directly extendable to other UN treaty bodies and future reporting cycles.

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